GRIT Interview Scheduler — Product Architecture Write-up

AI Product Engineer: Round 1 Assignment | NIAT Track Badge Certification System

Live Prototype: Render Cloud • SQLite • Python 3.11

1. Interview Journey State Machine (States & Transitions)

Every state a student's interview journey traverses from prerequisite completion to verified badge or revocation:

Ineligible
Prerequisites < 100%
Modules & quizzes pending

→

Eligible
Coursework 100% complete
Unbooked & ready to schedule

→

Hold (Locked)
Slot selected; 5-min timer
Lock prevents collisions

→

Scheduled
Booking confirmed; invite sent
Evaluator assigned in queue

→

Evaluation
Live 45-min technical call
Rubric & coding review

Post-Evaluation Outcomes & Terminal Integrity States:

● Passed
Badge unlocked & public certificate
Credential added to NIAT ledger

● Failed
Competencies not met
Detailed remediation notes

● Retake
Eligible after 7-day cooldown
Resets Track_Eligibility to 'Eligible'

● Nullified
Academic integrity violation flagged
Permanent credential revocation

TRANSITION TRIGGER MATRIX:
- Coursework Completed (100%) → Moves from Ineligible to Eligible (Unbooked).
- Click Open Slot → Atomically locks slot ('Locked') & starts 5-min client hold timer. Cancel/expire releases hold.
- Confirm Booking → Atomically writes Interview_Bookings, updates Slot to 'Booked', updates Student to 'Scheduled'.
- Interviewer Outcome Evaluation → 'Pass' updates to Passed; 'Fail' to Failed; 'Retake' crucially resets to Eligible.
- Admin Malpractice Flag → Overwrites student's Track_Eligibility to 'Nullified' from ANY state instantly.

2. Relational Data Model & Slot Collision Prevention

Schema entities, relational foreign keys, integrity constraints, and concurrency locking mechanisms:

Users

id (PK, int)
name (text)
email (text, UNIQUE)
role (Student | Interviewer | Admin)
created_at (datetime)

Track_Eligibility

id (PK, int)
student_id (FK → Users.id)
track_name (text)
status (Eligible|Scheduled|
Passed|Failed|Nullified)
updated_at (datetime)

Interviewer_Slots

id (PK, int)
interviewer_id (FK → Users.id)
start_time (datetime)
status (Available|Locked|
Booked)
created_at (datetime)

Interview_Bookings

id (PK, int)
slot_id (FK, UNIQUE → Slots.id)
student_id (FK → Users.id)
outcome (Pending|Pass|Fail|
Retake)
feedback (text)

HOW WE PREVENT TWO STUDENTS FROM BOOKING THE SAME SLOT (Three-Tier Defense):

1. Database Unique Constraint: Interview_Bookings defines slot_id as UNIQUE (slot_id INTEGER NOT NULL UNIQUE). Any concurrent attempt to insert a second bookin
2. Atomic ACID Transactions: The booking routine uses SQLite IMMEDIATE transactions (with conn:) executing a conditional write. The slot status is verified to be Avail
3. Interactive Booking Hold: When a student clicks a slot, the status is immediately transitioned to 'Locked'. Other students see the slot as unavailable in real time, pr

3. Edge Cases, Questions Handled & Product Assumptions

- Concurrent Booking Race Condition: Two students clicking the same slot simultaneously. Resolved via atomic status lock + UNIQUE constraint on slot_id.
- Hold Expiration / Abandonment: A student opens the modal but leaves the tab. Solved with a 5-minute countdown releasing 'Locked' slots back to 'Available'.
- Retake Cool-off Window: If evaluated as 'Retake', student is reset to 'Eligible', but rebooking is locked until a 7-day preparation window elapses (Nov 21).
- Malpractice Post-Issuance: A student passes their interview but is later caught cheating in contest code. Admin flag instantly revokes badge and locks dashbo
- Timezone Consistency: Interview slots store ISO timestamps and render in student's local timezone (IST) with unambiguous relative labels (e.g. 'Tomorrow 10
- Key Assumptions: Evaluators evaluate one candidate at a time; 45-minute standard interview duration; one active badge interview booking per track per stu

GRIT Interview Scheduler — Prototype & Interface Write-up

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4. The 3 Most Important Student Messages (Rendered Exactly As Prompted)

These verbatim messages provide immediate feedback across key milestones: booking confirmation, pass celebration, and retake guidance:

MESSAGE 1: BOOKING CONFIRMATION BANNER (Prompt 2)

“You’re booked! Your AI Engineering Track Interview is confirmed for Thursday, Nov 14 at 4:30 PM IST. A calendar invitation with your meet

MESSAGE 2: PASSED CELEBRATION CARD (Prompt 5 State 1)

“Congratulations! You have passed your Track Badge Interview. Your GRIT AI Systems Track Badge has been unlocked and verified on your
Interactive Feature: Clicking the link opens an authentic verifiable certificate modal with verification hash and public ledger link.

MESSAGE 3: RETAKE REQUIRED WARNING CARD (Prompt 5 State 2)

“Interview Outcome: Retake Required. Your interviewer has requested a retake on core architecture concepts. Your rebooking window will
Nullified Override: “Badge Nullified: Violation of academic integrity detected. All credentials have been revoked. ”

5. Before & After Screens & Working Prototype Architecture

How the interface conditionally evolves across the user lifecycle before and after an evaluation result:

BEFORE: Eligible (Unbooked)

- Header: 'AI Engineering Track: 100% Complete. You are eligible for your interview.'
- 100% Progress Dial + Verified checklist (4/4 modules complete).
- Available Slots Grid: Interactive cards with date, time, evaluator name, duration.
- Action: Clicking slot locks hold and launches confirmation modal.

BEFORE: Interview Scheduled

- Success Banner: Verbatim confirmation message with calendar dispatch.
- Scheduled Card: Displays confirmed slot date/time, interviewer, and Google Meet tag.
- Status Pill: Blue 'Scheduled' pill; open slots update to Booked.
- Evaluator Queue: Student automatically appears in evaluator's Daily Queue.

AFTER RESULT: Passed (Success)

- Celebratory Confetti Card with verified gold/emerald badge emblem.
- Verbatim congratulations message with interactive '[View Your Badge & Certificate]'.
- Certificate Modal: Official verified digital credential with verification ID.
- Available slots section cleanly hidden as requirement is satisfied.

AFTER RESULT: Retake / Nullified

- Retake: Amber warning card with verbatim message, 7-day cooldown date, & feedback n
- Nullified: Entire dashboard is OVERWRITTEN by full red locked screen (#4c0519).
- Revocation: 'Badge Nullified: Violation of academic integrity detected...'
- Complete lockout: Slot booking disabled, case reference displayed.

6. Live Working Prototype & Production Verification

Hosted online with production WSGI (Render Cloud) with complete automated test coverage:

- Production Platform: Render Cloud Web Service (Gunicorn WSGI + Flask 3.0.3) • Repository: GitHub
- Automated Test Suite: 23 out of 23 Unit & Integration Tests Passing (0 failures, 100% green coverage):
  1. test_db.py (9 tests: schema constraints, cascading deletes, foreign keys, unique emails, slot dates)
  2. test_booking_flow.py (4 tests: hold locking, booking transactions, student status to Scheduled, exact message)
  3. test_interviewer_outcomes.py (5 tests: availability manager, queue listing, Pass/Fail/Retake status mapping)
  4. test_admin_and_dynamic_states.py (5 tests: cohort metrics, malpractice flag to Nullified, Passed/Retake/Nullified messages)